$$\rho_{aa'} = \langle a|\Psi\rangle\langle\Psi|a'\rangle,$$

or

$$\rho_{aa'} = \psi(a)\psi^*(a').$$

7.5 ENTANGLEMENT FOR TWO SPINS

Before leading you further into the world of entanglement, I'll give you a simple definition and a quick warm-up exercise. If Alice only has a single spin in a known state, her density matrix is defined to be

$$\rho_{aa'} = \psi^*(a')\psi(a).$$

This equation tells you how to calculate an element of Alice's density matrix. If we stick with our familiar σ_z basis, each index a and a' can take the values *up* and *down*, so Alice has a 2 × 2 density matrix.

Exercise 7.4: Calculate the density matrix for

$$|\Psi\rangle = \alpha|u\rangle + \beta|d\rangle.$$

Answer:

$$\psi(u) = \alpha; \qquad \psi^*(u) = \alpha^*$$
$$\psi(d) = \beta; \qquad \psi^*(d) = \beta^*$$

$$\rho_{a'a} = \begin{pmatrix} \alpha^*\alpha & \alpha^*\beta \\ \beta^*\alpha & \beta^*\beta \end{pmatrix}.$$